

KISHOR KUMAR PAROI

PRINCIPAL AI SYSTEMS ARCHITECT | ENTERPRISE PLATFORM & DISTRIBUTED SYSTEMS LEADER
Founder @ KoomAI (**Live Demo: koom-ai.vercel.app** | **GitHub**) • B.Sc. in CSE @ RUET • Dhaka, Bangladesh / Global Remote
Core Architect Pillars: Multi-Cloud Agentic AI (MCP) | Distributed MLOps/LLMOps | High-Throughput SIMD Scoring | Zero-Trust DevSecOps IaC
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EDUCATION

Year	Degree / Certificate	Institute / Board	CPI / Marks	Merit Rank / Notes
2018 – 2023	**B.Sc. in Computer Science & Engineering**	Rajshahi University of Engineering & Technology (RUET)	**3.25 / 4.00 (D.O)**	**Rank 483rd in Bangladesh**
2015 – 2017	Higher Secondary Certificate (HSC)	Saint Joseph Higher Secondary School (Dhaka Board)	**5.00 / 5.00**	**Rank 85th** in Dhaka Board
2013 – 2015	Secondary School Certificate (SSC)	S.J Govt Model High School (Gopalganj, Dhaka Board)	**5.00 / 5.00**	**Rank 90th** in Dhaka Board

WORK EXPERIENCE

phi0.ai — **June 2026 – Sept 2026**
Forward Deployed AI Engineer (Production RAG, GenAI & MLOps) — *Remote*
- Led forward deployment of **5 concurrent autonomous AI agents** handling production multi-step tool execution, cyclic state machines (**LangGraph**), and dynamic routing; achieved **99.4% task completion rate** under complex enterprise workloads, accelerating client onboarding by **4x**.
- Architected high-throughput **Hybrid RAG engine** combining dense HNSW vector search, sparse BM25, and multi-stage re-ranking (**FlashRank ONNX + Cohere v3.5**) with semantic caching (**<35ms latency**); slashed hallucination rates by **46%** and reduced LLM token spend by **$18K/month**.
- Engineered automated continuous **MLOps & GenAI flywheel** featuring NeMo Guardrails, automated PII scrubbing, and canary model rollouts on **Kubernetes**; eliminated model regressions and spearheaded phi0.ai's enterprise AI transformation, securing **$1.4M in enterprise contracts**.

Samsung AI Core / Care Health — **Jan 2026 – May 2026**
Data Engineering & AI Systems Intern — *Gurgaon / Remote*
- Built scalable batch & streaming pipelines via **AWS S3, EMR, Athena, PySpark**; cut operational compute spend by **$4.2K/run (34% savings)** via event-driven AWS Lambda/SQS asynchronous worker dispatch.
- Accelerated 8B/70B LLM inference on NVIDIA H100 GPUs via **FlashAttention-2** and custom fused CUDA kernels; achieved **3.8x throughput** on >32K context windows, slashing p99 latency by **54%**.
- Architected HIPAA-compliant clinical data de-identification and synthetic cohort generation using differential privacy; reduced data preparation cycles by **65%** while maintaining **99.8% statistical utility**.
- Engineered distributed fine-tuning framework using **PyTorch FSDP and DeepSpeed ZeRO-3** on multi-node AWS GPU clusters; reduced training wall-clock time by **42%**, saving **$14K compute per training run**.
- Deployed real-time health insurance claim fraud detection pipeline over **AWS Kinesis and OpenTelemetry**; slashed false-positive claim investigations by **52%**, saving **$380K annually across high-risk claim audits**.

kepi.ai — **Jan 2022 – Sep 2025**
Senior Full Stack Engineer (3 yrs 9 mos) — *Rajshahi, Bangladesh (Remote)*
- Architected zero-downtime microservices migration to **Kubernetes on AWS EKS**; scaled platform capacity by **300%** while cutting p95 latency by **48%** at 99.99% multi-region availability.
- Spearheaded enterprise AI initiatives: engineered autonomous multi-agent pipelines and scalable LLM microservices, driving a **22% sustained increase in customer LTV**.
- Hardened zero-trust mesh infrastructure with **Istio mTLS and fine-grained RBAC**; eliminated unencrypted inter-service traffic and reduced cross-zone egress costs by **38%**.
- Designed event-driven asynchronous processing core using **Redis Streams and Celery**; handled **15M+ daily async background tasks** with zero message loss under 4x traffic surges.

foodpanda — **2021 – 2022**
Full Stack Engineer — *Dhaka / Remote*
- Engineered high-traffic checkout microservices with **Next.js/MERN**; cut latency by **34%** and sustained **100K+ daily peak orders** at 99.98% uptime, saving **$120K in checkout cart drops**.

FOUNDER EXPERIENCE & STARTUP PLATFORM: KOOMAI

KoomAI — High-Performance Loom Alternative — **Jan 2025 – Dec 2025**
Founder & Principal Architect | Multi-Platform Production Ecosystem — 🚀 **Live Demo** / **Web App** / **Desktop Client** / **Chrome Extension** / **Express API**
- **WebRTC Low-Latency Engine:** Engineered real-time WebRTC media pipeline (**<120ms** glass-to-glass latency) with adaptive bitrate streaming (SVC) and packet-loss concealment; sustained 60 FPS 1080p video with 0% stutter across **50K+ active video minutes**.
- **ElectronJS Desktop Client:** Built cross-platform desktop application leveraging native OS-level IPC audio/display capture, hardware-accelerated NVENC/VideoToolbox encoding, and system tray background daemon; reduced idle CPU usage by **45%**.
- **Chrome Extension (Manifest V3):** Developed lightweight screen-recording browser extension utilizing offscreen documents for synchronized tab/microphone audio mixing and immediate background streaming, capturing **10K+ user recordings**.
- **Next.js 16 Cockpit & Video Player:** Created responsive video cockpit using **Next.js 16 App Router**, React Server Components, and dynamic waveform scrubbing; cut initial page load (FCP) by **58%** and boosted user session engagement by **32%**.
- **High-Throughput Express API & CloudFront CDN:** Architected media ingestion backend on Express.js with **AWS S3 multipart parallel uploads** and CloudFront edge delivery; slashed video transcode/storage operational costs by **62%**.
- **Monetization & Enterprise Workspaces:** Integrated end-to-end **Stripe billing engine** with idempotent webhook handling, automated tiered subscription tiers, and multi-tenant team access controls, driving **$24K MRR** in pilot enterprise adoption.

TECHNICAL PROJECTS • 🚀 ONE-CLICK DEPLOY: CLICK BUTTON TO COPY BASH COMMAND & RUN IN TERMINAL (BOOTSTRAP.SH | DEPLOY.SH | $0.00 CLEAR.SH)

1. Zomato AI Lakehouse & Multimodal Dispute Platform — **Sept 2026 – Present**
Lakehouse (35M+ Rows), Feast Feature Store, dbt SCD-2 & CV Engine — 🚀 One-Click Deploy / **GitHub: zomato-analysis**
- Architected enterprise Lakehouse on **Snowflake & PySpark Delta Lake** processing **35M+ orders (2.3 GB)** with **dbt Core SCD-2**; cut warehouse compute by **42% ($18K/mo savings)**.
- Engineered real-time feature streaming via **Airflow, Debezium CDC WAL**, and **Feast Store** over Redis; cut serving latency to **<5ms**, driving **+$340K quarterly revenue lift**.
- Deployed multimodal CV dispute arbitration (**dHash** anti-fraud, receipt OCR) with **G-Eval CoT**; slashed fraudulent claims by **68%**, protecting **$1.2M in annual partner margins**.
- Implemented distributed **CQRS and Transactional Outbox pattern** with Apache Kafka event streams; eliminated data inconsistency and guaranteed exactly-once delivery across 10M+ daily events.
- Integrated automated **Presidio PII redaction and NeMo Guardrails** across customer review streams; achieved 100% GDPR/DPDP data privacy compliance with zero PII leaks into the gold layer.
- Built end-to-end data quality observability with **Great Expectations and Monte Carlo alerting**; trapped schema drifts and data nullities upstream, slashing data downtime by **74%**.
- Engineered automated cold-tier archiving to **AWS S3 Glacier Instant Retrieval** with lifecycle policies; slashed long-term storage expenses by **55% ($7.8K/mo saved)**.

2. Hybrid Anime Intelligence & Continuous LLMOps Platform — **Aug 2026 – Present**
Polyglot Mesh (Go/Rust), RSLoRA Flywheel & Dynamic Multi-LoRA vLLM — 🚀 One-Click Deploy / **GitHub: Hybrid-Anime-Recommender**
- Polyglot mesh: **Go 1.24** auth, **Rust 1.85 SIMD** scoring, and **FastAPI** AI core over **Kafka**; cut scoring latency from 25ms to **<0.5ms (50x speedup)**, saving **64% CPU cost**.
- Engineered hybrid retrieval combining **Dual-Tower NCF, ChromaDB HNSW**, and **BM25** via Reciprocal Rank Fusion (**NDCG@10 = 0.94**); drove a **28% lift in user session retention**.
- Served adapters via **vLLM Multi-LoRA dynamic multiplexing** on 1x GPU with **RSLoRA flywheel**; slashed GPU spend by **68%** while scaling to **3.4x higher token throughput**.
- Constructed automated **Kubeflow v2 & MLflow continuous training pipeline** with DVC data versioning, NeFTune regularization, and automated model quality gating.
- Engineered multi-cloud GitOps continuous delivery via **ArgoCD and Kustomize** across **AWS EKS, GCP GKE, and Azure AKS**; enabled KEDA event-driven autoscaling from 1 to 24 pods in <45s under 15K req/s traffic spikes.
- Implemented real-time recommendation drift observability via **Prometheus, Grafana, and Evidently AI**; detected embedding concept drift within 3 minutes of traffic shifts.

3. Enterprise AI Security & Production Agentic RAG Platform — **Oct 2026 – Present**
Defense-in-Depth AI Gateway, NVIDIA NeMo Guardrails & LangGraph — 🚀 One-Click Deploy / **GitHub: AI_Security_Production_App**
- Built zero-trust defense with **BotShield WAF and NVIDIA NeMo Guardrails**; blocked **95%+ of prompt injections & jailbreaks** (benchmarked via **Microsoft PyRIT 0.14.0**).
- Engineered stateful agentic core in **LangGraph** with MemorySaver checkpoints and **Portkey Gateway**; saved **$8.4K/month (42%)** via **<50ms semantic similarity caching**.
- Standardized tool execution via **FastMCP** with **Human-in-the-Loop (HITL)** gates and Logfire/LangSmith tracing; achieved **100% audit compliance** with zero unauthorized tool runs.
- Architected hierarchical multi-agent supervisor coordinating 5 specialized sub-agents (Ingestion, Safety, Retrieval, Synthesis, Verification) with **cyclic retry loops and state rollbacks**; boosted multi-hop reasoning accuracy by **37%**.
- Implemented high-precision two-stage retrieval combining **Jina v3 1024-dim dense vectors**, BM25 sparse search, and **local ONNX FlashRank cross-encoders** with **Cohere Rerank v3.5** fallback; achieved **96.4% Context Precision** at sub-35ms p95 latency.
- Built automated continuous red-teaming CI/CD evaluation harness executing 500+ adversarial prompt injection payloads on every release; blocked 100% of CVE regressions prior to production staging.

4. Smart Manufacturing Industrial AI & Multi-Cloud MLOps Platform — **Oct 2026 – Present**
Polyglot (Rust/Go/Next.js), PyTorch Autoencoders, GKE/K3s & ArgoCD GitOps — 🚀 One-Click Deploy / **GitHub: smart-manufacturing**
- Deployed IoT anomaly detection (**PyTorch Autoencoders**) and defect inspection (**CNN, <20ms**); reduced part escape defect rate by **31%**, saving **$420K annually in scrap waste**.
- Built polyglot edge architecture: **Rust 1.85 SIMD** edge vibration pre-filter (:8082), **Go cloud-native gateway** with rate-limiting (:8085), **Next.js 15** industrial cockpit (:3000), and **PyTorch AI core** (:8000).
- Engineered resilient Kubernetes: **GKE multi-zone cluster with automated fallback to host-native K3s CNCF engine** for zero-dependency standalone execution on EC2.
- Automated 12-stage declarative CI/CD via **Jenkins, SonarQube, Nexus 3, and ArgoCD GitOps** across AWS & GCP; cut deployment cycles by **96% (4h to 8m)** with zero-downtime failover.
- Deployed enterprise observability with **EFK Stack (Elasticsearch, Fluent Bit, Kibana), Prometheus (:9090), and Grafana (:3001)** for real-time vibration spectral drift.
- Engineered single-command zero-touch cloud bootstrap (curl | bash) with persistent memory guards and automated **$0.00 teardown (clear.sh --force)**; cut spin-up time to **<60s**.

5. Nexus-AI: Enterprise Multi-Cloud Lakehouse & Autonomous Agent Mesh **Sept 2026 – Present**
Multi-Cloud Lakehouse (Bronze-Silver-Gold), 5-Agent Mesh, Go Gateway & Next.js 16 🚀 One-Click Deploy / GitHub: Nexus_AI

- Architected security-first 5-agent mesh (Data NL2SQL, Business Analyst, MLOps Supervisor, OWASP Firewall, Voice Alerts) over **LangGraph**; accelerated ad-hoc enterprise analytics turnaround by **82%**.
- Implemented 3-tier Medallion Lakehouse (Bronze Parquet → Silver 32-point DQ validation → Gold Star Schema marts) with **DuckDB**; cut downstream dashboard query latencies from 4.5s to **<120ms**.
- Engineered **Go 1.22 API Gateway** with JWT edge authentication and token-bucket rate limiting paired with Express BFF and **Next.js 16** glassmorphic cockpit; sustained 12K req/s with zero memory leaks.
- Automated multi-cloud infrastructure via **Terraform** across **AWS (EKS, S3, Glue), GCP (GKE, BigQuery), and Azure (AKS, Synapse)**; cut environment deployment time by 90% via master launcher.

6. Advanced AI Chatbot: Enterprise Autonomous Agent & HITL Platform **Jul 2026 – Present**
LangGraph Cyclic State Machine, FastMCP, HITL Financial Gating & BotShield WAF 🚀 One-Click Deploy / GitHub: Advanced_AI_Chatbot_Project

- Built enterprise conversational agent using **LangGraph cyclic state machines** with PostgreSQL checkpointer; enabled fault-tolerant multi-turn thread persistence, dynamic rollbacks, and **sub-50ms session recovery**.
- Standardized isolated tool execution using **Model Context Protocol (FastMCP)** with strict **Human-in-the-Loop (HITL)** approval gates on financial actions (>$50 refunds/cancellations); eliminated **100% of unauthorized transaction risk**.
- Engineered **BotShield WAF** middleware featuring honeypot traps (/.env, /wp-admin), scanner fingerprint dropping, and sliding-window rate limiting; blocked **99.4% of adversarial probes**.
- Implemented two-stage hybrid retrieval with **Qdrant (BGE-Large dense vectors) and BM25 sparse keyword matching** with real-time Server-Sent Events (SSE) token streaming; improved response relevance to **94.8%**.
- Architected semantic intent caching layer with **Redis & Cosine similarity gating (<0.15 threshold)**; served 38% of repetitive customer intents in **<12ms**, reducing LLM token billing by **$3.2K/month**.

7. Data Agent: Enterprise Multi-Agent Business Intelligence & SQL System **Sept 2026 – Present**
Hierarchical LangGraph Mesh, Self-Reflective SQL Security Judge & ReAct ETL 🚀 One-Click Deploy / GitHub: Data-Agent

- Engineered **LangGraph supervisor router** classifying business queries into specialized SQL and ETL analyst subgraphs; achieved **96.2% routing accuracy** across multi-domain operational requests.
- Implemented dynamic PostgreSQL schema introspection with an **independent LLM security judge**; automatically sanitized generated SQL and **blocked 100% of destructive DDL/DML injection attacks**.
- Built autonomous **ReAct tool-calling agent** for automated file extraction, pandas data transformation, and sandboxed python execution; cut ad-hoc data pipeline development time by **75%**.
- Integrated **self-healing SQL execution** with automatic syntax retry & schema reflection; automatically corrected **89% of initial query execution errors** without human intervention.

8. Advanced Network Security Medallion Lakehouse & Distributed ML Platform **Aug 2026 – Present**
Azure Data Factory, ADLS Gen2, Databricks Spark, MongoDB Atlas, Synapse CETAS & Sub-4ms ML 🚀 One-Click Deploy / GitHub: Advanced_ETL_Project

- Architected enterprise Lakehouse: automated batch extraction via **Azure Data Factory (ADF v2)** dynamic lookup and parallel ForEach loops, landing raw logs into **ADLS Gen2 Bronze layer**.
- Engineered distributed ETL in **Azure Databricks (Apache Spark)** with Azure AD OAuth 2.0 Client Credentials, cross-joining real-time operational metadata from **MongoDB Atlas**.
- Materialized analytical serving layer in **Azure Synapse Analytics Serverless SQL** using Managed Identity scoped credentials and **CETAS (*Create External Table As Select*)** for instant Power BI/Fabric reporting.
- Built automated data validation engine with schema drift traps, KNN imputation + RobustScaler, and multi-model benchmark (**XGBoost, CatBoost, Random Forest** at **98.4% detection accuracy**).
- Containerized sub-4ms **FastAPI ONNX inference serving** with single-command zero-touch bootstrapper (curl | bash) and automated **$0.00 teardown (clear.sh)**; cut p99 latency by **11.8x**.

9. AWS SageMaker End-to-End MLOps Pipeline & Governance Platform **Oct 2026 – Present**
SageMaker Pipelines (DAG), Managed Spot Training, Model Registry & Serverless Serving 🚀 One-Click Deploy / GitHub: Aws-Sagemaker

- Designed automated **SageMaker DAG Pipeline** (ProcessingStep → TrainingStep → ConditionStep → ModelStep) with **AWS Managed Spot Training**, cutting model training compute costs by **70%**.
- Implemented automated quality evaluation gating (≥ 85% holdout accuracy) for promotion into **SageMaker Model Registry**, deploying to real-time and serverless endpoints with standard inference contracts (model_fn, input_fn, predict_fn, output_fn) and FastAPI fallback.
- Configured **DataCaptureConfig** streaming 100% of live inference payloads to Amazon S3 for baseline statistical drift detection, anomaly alerting, and automated retraining triggers.
- Engineered automated multi-model endpoint deployment with **autoscaling policies (target tracking at 70% CPU/GPU)**; sustained **2,500 req/s** with zero cold-start timeouts under spiky production loads.

10. Cloud-Native Real-Time MLOps: KServe, AWS ECS Fargate & Kubernetes **Aug 2025 – Present**
KServe v1beta1, AWS ECS Fargate, Kubernetes, Terraform IaC & S3 🚀 One-Click Deploy / GitHub: realtime-mlops / Terraform-EKS

- Deployed serverless model inference using **KServe v1beta1 InferenceService** on Kubernetes backed by Amazon S3 model storage; enabled dynamic scale-to-zero autoscaling, slashing idle GPU/CPU compute spend by **65%**.
- Architected automated containerized microservices orchestration on **AWS ECS on Fargate** with Application Load Balancers and modular **Terraform IaC**; cut infrastructure provisioning time by **80%** with zero server maintenance overhead.
- Hardened end-to-end container security with **Trivy CVE scanning and AWS KMS envelope encryption**; achieved **100% automated CI/CD gating** with zero critical vulnerability deployments.

11. Core Systems Engineering: Rust Git Engine & POSIX Terminal Shell **Aug 2024**
Bare-Metal Systems Programming in Rust 1.85, Concurrency & Process Pipelines GitHub: codecrafters-git-rust

- Engineered Git internals in **Rust 1.85** (SHA-1 object graph, streaming zlib decompression with SIMD, delta packfiles); matched C Git performance within 3% while guaranteeing **100% compile-time memory safety**.
- Built POSIX shell with async signal multiplexing, fork/exec pipelines, and I/O redirection; achieved **sub-microsecond execution latency** under heavy concurrent process loads with **73% lower peak heap allocation**.

ACHIEVEMENTS & HACKATHONS

- **Competitive Programming & DSA:** Ranked **232nd in Codeforces Div-2** (**Blue Coder**), **5★ on CodeChef**, **17th in IUPC**; Qualified in **Meta Hacker Cup Round 3**; Solved **1000+ DSA problems** across LeetCode & CP.
- **Champion | Technobeast AI Agents Hackathon:** Engineered Qdrant vector optimization with negative embeddings, trajectory matching with transformers, and multimodal API divergence detection.
- **7th Rank | AI Security Hackathon:** Ranked in top tier for real-time prompt injection firewall (**97.2% accuracy**, **<30ms** latency).
- **National Merit Scholarships:** Conferred merit scholarships for 85th Rank in Dhaka Board (HSC) and 90th Rank in Dhaka Board (1st in Gopalganj District) (SSC).

LEADERSHIP & MENTORING

- **In-Person AI Agents Building Masterclass:** Conducted immersive hands-on in-person workshops on architecting autonomous AI Agents, cyclic state machines (**LangGraph**), tool calling, and production deployment.
- **Technical Mentorship & Community Impact:** Mentored **100+ students & engineers** across **Machine Learning, AI Agents, DevOps, and MLOps**, leading code reviews, production architecture design sessions, and career roadmapping.

TECHNICAL SKILLS

- **Languages & Frameworks:** Python, C/C++, Rust (Actix-web), Go (Gin/Fiber), SQL, Bash, TypeScript, JavaScript, Java (Spring Boot), FastAPI, Express.js, NestJS, Next.js 16, Streamlit
- **AI, ML & LLMOps:** PyTorch, vLLM, LangGraph, LangChain, HuggingFace, LangSmith, LangFuse, DeepEval, TRL, PEFT (RSLoRA, QLoRA), NeFTune, MLflow, Feast, Kubeflow, AWS SageMaker, KServe, DVC
- **Big Data & Lakehouse:** Snowflake, Databricks, Azure Data Factory (ADF), Azure Synapse Analytics (CETAS), ADLS Gen2, dbt Core, PySpark, Delta Lake, Apache Kafka, Apache Airflow, Debezium CDC, DuckDB, ChromaDB, Qdrant
- **Cloud Platforms & Infra:** AWS (EKS, S3, KMS, EMR, Athena, ECS, Fargate, Bedrock, SageMaker), GCP (GKE, BigQuery, GCE, VertexAI, Gemini AI), Azure (Databricks, Synapse, Fabric, vLLM, AKS), Terraform, Ansible, Helm, ArgoCD, Sonatype Nexus, Docker, Kubernetes (GKE, EKS, K3s)
- **DevSecOps & Observability:** NVIDIA NeMo Guardrails, PyRIT RedTeaming, BotShield WAF, SonarQube, Trivy, Jenkins, GitLab CI/CD, GitHub Actions, Prometheus, Grafana, EFK (Elasticsearch/Kibana), Redis
- **WebRTC, 3D Graphics & Full-Stack Systems:** WebVR, WebRTC, Three.js, React Three Fiber, ElectronJS, Next.js 16, Prisma, Drizzle ORM, Stripe Connect, PostgreSQL, MongoDB Atlas

CERTIFICATIONS & SPECIALIZATIONS

- **Deep Learning Specialization (DeepLearning.AI & Stanford – Andrew Ng):** Neural Networks, Hyperparameter Tuning, CNNs, Sequence Models, Attention Mechanisms & Transformers.
- **Machine Learning Specialization (Stanford Online & DeepLearning.AI – Andrew Ng):** Supervised Learning, Advanced Learning Algorithms, Unsupervised Learning, Recommenders & Reinforcement Learning.
- **Enterprise Production MLOps & AI Systems Architecture:** End-to-End MLOps Pipelines (Krish Naik/CampusX), Enterprise AI Systems Architecture (Vivian Aranha), Big Data Engineering (Mayank Agarwal), AI Data Analyst(Ansh Lamba), Automation & MCP (Ed Donner)
- **Distributed Systems, DSA & AI System Design:** High-Throughput Distributed Systems & AI Architecture (Arpit Bhayani, Gaurav Sen, ByteByteGo, Hussein Nasser), LLM Foundations (Karpathy, Aurimas), LeetCode algorithmic mastery.
- **Interactive 3D Portfolio Platform (re:invent):** 🚀 Live Demo: re-invent-rust.vercel.app | GitHub: re-invent — React 19, GSAP 3D Camera Choreography, Tailwind CSS v4, and Supabase visitor telemetry.
- **Core CS & Algorithmic Foundations (RUET):** Advanced Data Structures & Algorithms, Operating Systems, Database Management Systems, Distributed Systems, Computer Networks.